

P–SV–SH Reciprocity: Mode Conversion, Flux Normalisation and the T-Matrix Metric

T. M. Nestor

Research School of Earth Sciences, Australian National University

Status. Explanatory note. The far-field relation (4) is validated numerically for the P↔SV channel of the exact Mie sphere (Section 2.1; Python only, no Mathematica cross-check yet). The flux normalisation (6) and the T-matrix metric statement (7) remain *unvalidated* in this repository.

1 Origin: Betti–Rayleigh reciprocity

Let the medium be linear elastic with the major symmetry $c_{ijkl} = c_{klij}$. It may be anisotropic, heterogeneous or attenuating (Betti, 1872; Strutt, 1873, 1894; Achenbach, 2003). For two time-harmonic states A and B in the same medium, with no sources inside a closed surface S ,

$$\oint_S (\mathbf{u}^A \cdot \mathbf{t}^B - \mathbf{u}^B \cdot \mathbf{t}^A) dS = 0, \quad \mathbf{t} = \boldsymbol{\sigma} \cdot \hat{\mathbf{n}}. \quad (1)$$

Two properties of (1) matter below.

- It is **bilinear**: there is no complex conjugate. It is not an energy statement, so it survives attenuation.
- It is **antisymmetric** in $A \leftrightarrow B$, so it is a symplectic form on displacement–traction pairs $(\mathbf{u}, \mathbf{t})$.

The point-source form is the Green’s-tensor symmetry

$$G_{ij}(\mathbf{x}, \mathbf{x}') = G_{ji}(\mathbf{x}', \mathbf{x}), \quad (2)$$

which swaps source with receiver and force component with receiver component (Knopoff and Gangi, 1959); see Aki and Richards (2002, §2.3).

2 Mode conversion breaks naive symmetry

For scalar waves the far-field amplitude obeys $f(\hat{\mathbf{x}}; \hat{\mathbf{d}}) = f(-\hat{\mathbf{d}}; -\hat{\mathbf{x}})$. Elastic waves have two speeds, α (P) and β (S), and the converted amplitudes are *not* symmetric in displacement units.

Write the scattered far field for a unit-amplitude incident plane wave of direction $\hat{\mathbf{d}}$ and polarisation $\hat{\mathbf{p}}$ as $\mathbf{f}_{\text{out} \leftarrow \text{in}} e^{ik_{\text{out}} r}/r$. The far-field Green's tensor carries a mode-dependent factor $1/(4\pi\rho c^2)$ (Aki and Richards, 2002, §4.2):

$$\mathbf{G}^P \sim \frac{\hat{\mathbf{x}}\hat{\mathbf{x}}}{4\pi\rho\alpha^2} \frac{e^{ik_P r}}{r}, \quad \mathbf{G}^S \sim \frac{\mathbf{I} - \hat{\mathbf{x}}\hat{\mathbf{x}}}{4\pi\rho\beta^2} \frac{e^{ik_S r}}{r}. \quad (3)$$

Building each plane wave as the limit of a distant point force and applying (2) to the scattered part gives

$$k_{\text{in}}^2 \hat{\mathbf{q}} \cdot \mathbf{f}_{\text{out} \leftarrow \text{in}}(\hat{\mathbf{x}}; \hat{\mathbf{d}}, \hat{\mathbf{p}}) = k_{\text{out}}^2 \hat{\mathbf{p}} \cdot \mathbf{f}_{\text{in} \leftarrow \text{out}}(-\hat{\mathbf{d}}; -\hat{\mathbf{x}}, \hat{\mathbf{q}}). \quad (4)$$

This is the elastic form of the reciprocity relation for scattering amplitudes (Pao and Varatharajulu, 1976; Varatharajulu, 1977). Consequently:

- **P→P and S→S (including SV↔SH).** Equal speeds on both sides, so (4) is plain symmetry. The only subtlety is the sign convention of the SV unit vector under direction reversal.
- **P→S versus S→P.** These differ by $(k_S/k_P)^2 = (\alpha/\beta)^2$, which is 3 for a Poisson solid.

2.1 Check against the exact Mie sphere

For a sphere, rotational symmetry reduces (4) to a statement about a single pair of Mie amplitudes. Take P incident along $+\hat{\mathbf{z}}$ and observe SV at polar angle θ in the xz -plane, projected on $\hat{\boldsymbol{\theta}}$; call this $f_{P \rightarrow SV}(\theta)$. The reverse experiment has S incident along $-\hat{\mathbf{x}}(\theta)$ with polarisation $\hat{\boldsymbol{\theta}}$, and P observed along $-\hat{\mathbf{z}}$. A rotation about $\hat{\mathbf{y}}$ by $\pi - \theta$ carries it onto SV incident along $+\hat{\mathbf{z}}$ with polarisation $-\hat{\mathbf{x}}$, observed at $(\theta, \phi = \pi)$. The $\cos \phi$ dependence of the radial P field cancels the polarisation sign, and projecting the outgoing P wave along $-\hat{\mathbf{z}}$ onto $\hat{\mathbf{z}}$ leaves one minus sign:

$$k_P^2 f_{P \rightarrow SV}(\theta) = -k_S^2 f_{SV \rightarrow P}(\theta), \quad (5)$$

where $f_{SV \rightarrow P}(\theta)$ is the radial P amplitude for an $\hat{\mathbf{x}}$ -polarised SV wave incident along $+\hat{\mathbf{z}}$.

The gate

`scripts/gate_mie_psv_reciprocity.py`

evaluates both sides using `compute_elastic_mie` and `mie_far_field`. The P- and SV-incident branches use different right-hand sides and different far-field assembly (the $m = 0$ Legendre expansion versus the renormalised $m = 1$ N-type expansion), so a normalisation fault in either branch would break (5). The background is $\alpha = 5$ km/s, $\beta = 3$ km/s, $\rho = 2.5$ g/cm³, with $(\alpha/\beta)^2 = 2.78$. The gate covers $k_S a \in \{0.1, 0.5, 1.5, 4.0\}$, 60 angles in $[0.05, \pi - 0.05]$, and three contrasts: *moderate* ($\Delta\lambda = +2$ GPa, $\Delta\mu = +1$ GPa, $\Delta\rho = +100$ kg/m³), *density only* ($\Delta\rho = +250$ kg/m³) and *strong soft* ($\Delta\lambda = -8$ GPa, $\Delta\mu = -10$ GPa, $\Delta\rho = -500$ kg/m³).

Contrast	$k_S a = 0.1$	0.5	1.5	4.0
Moderate	9.3×10^{-5}	1.1×10^{-5}	2.8×10^{-6}	5.1×10^{-6}
Density only	5.9×10^{-5}	5.1×10^{-6}	1.9×10^{-6}	3.2×10^{-6}
Strong soft	1.1×10^{-4}	1.4×10^{-5}	6.1×10^{-6}	5.6×10^{-6}

The table gives the maximum deviation over angle, normalised by the peak amplitude. The residual comes from `mie_far_field` evaluating the field at $r = 10^6 a$ rather than at infinity, an $\mathcal{O}(n^2/kr)$ near-field remainder that is largest at low frequency. Two negative controls fail in every case, as they must. The unweighted scalar form $f_{P \rightarrow SV} = -f_{SV \rightarrow P}$ is off by exactly $(\alpha/\beta)^2 - 1 = 1.78$, and reversing the sign of (5) gives an error of 2.

3 Flux normalisation restores symmetry

Normalise each amplitude so that its squared modulus is an energy flux. The incoming and outgoing spherical-wave amplitudes carry $\sqrt{\rho c}$, and the plane-wave expansion contributes $1/k_{\text{in}}$, giving

$$\tilde{\mathbf{f}}_{\text{out} \leftarrow \text{in}} = \frac{k_{\text{in}}^{3/2}}{k_{\text{out}}^{1/2}} \mathbf{f}_{\text{out} \leftarrow \text{in}}. \quad (6)$$

In these units the scattering matrix over (P, SV, SH) $\times$ directions is **symmetric** under direction reversal, $\tilde{\mathbf{S}} = \tilde{\mathbf{S}}^T$, and in the absence of attenuation it is also unitary (Varatharajulu and Pao, 1976; Varatharajulu, 1977). Reciprocity supplies the symmetry and energy conservation the unitarity. They are independent facts that share one normalisation.

The layered-media analogue is given by Aki and Richards (2002, Ch. 5) and Kennett (1983): the displacement coefficients satisfy $R_{PS} \neq R_{SP}$, whereas the energy-normalised coefficients, with a factor $\sqrt{\rho c \cos \theta}$ per wave, form a symmetric matrix.

4 Decoupling of SH

In two-dimensional or laterally homogeneous geometries (plane layers, spheres) SH decouples from P–SV, and reciprocity splits into blocks:

- **SH block:** scalar with a single speed, so plainly symmetric;
- **P–SV block:** 2×2 . The $P \leftrightarrow SV$ off-diagonal terms need the weighting of (6).

For a sphere in the vector spherical-wave basis (Morse and Feshbach, 1953; Ben-Menahem and Singh, 1981):

Basis field	Character	Wave type	Coupling
L	longitudinal	P	couples with N
N	poloidal	SV	couples with L
M	toroidal	SH	decoupled

Within each multipole order n , $\mathbf{L}$ and $\mathbf{N}$ form the P–SV block and $\mathbf{M}$ stands alone.

5 T-matrix form and the reciprocity metric

In a basis of regular and outgoing fields, (1) becomes the matrix statement (Waterman, 1976; Pao, 1978)

$$\mathbf{T}^\top W = W \mathbf{T}, \quad (7)$$

where W is the Gram matrix of the bilinear form $\oint(\mathbf{u}^A \cdot \mathbf{t}^B - \mathbf{u}^B \cdot \mathbf{t}^A) dS$ between regular and outgoing basis fields. That form is the symplectic J acting on displacement–traction columns.

- With un-normalised $\mathbf{L}/\mathbf{M}/\mathbf{N}$ columns, W is diagonal with mode-dependent entries (k_p for $\mathbf{L}$, k_s for $\mathbf{M}$ and $\mathbf{N}$, together with ρ and μ factors). $\mathbf{T}$ is then only W -symmetric, not symmetric.
- Rescaling the columns by $W^{1/2}$ yields $\tilde{\mathbf{T}} = W^{1/2} \mathbf{T} W^{-1/2}$ with $\tilde{\mathbf{T}} = \tilde{\mathbf{T}}^\top$ exactly: the spherical-wave counterpart of (6).

This is how two operators built with different column normalisations (T0 and G0 in the IntraPlane Foldy–Lax work) can each be reciprocal yet appear to disagree: each is symmetric only in its own metric. Evaluating W from the explicit displacement–stress arrays with the plain symplectic J checks both against the same form, so any residual mismatch is a genuine error rather than a normalisation artefact.

Numerical check. Build W from the explicit arrays at a test radius and confirm, separately for the P–SV and SH blocks, that

$$\frac{\|\mathbf{T}^\top W - W \mathbf{T}\|}{\|W \mathbf{T}\|} \approx \varepsilon_{\text{mach}}. \quad (8)$$

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